

Sai Ravi Teja Gangavarapu

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EDUCATION

Carnegie Mellon University

Master of Software Engineering - Scalable Systems (ML Systems, Statistics, QA, Formal Methods)

Pittsburgh, PA

Aug 2025 – Dec 2026

University of Florida

Senior Cert + Master's coursework in Computer Science (Algorithms, UX, Graphics - 4.0)

Gainesville, FL

Jan 2024 – May 2024

SKILLS

C++, Python, ROS2, motion planning & control (MPPI), simulation & sim-to-real, state estimation; PyTorch, deep learning, perception (segmentation, LiDAR/RGB-D); multi-agent systems, CUDA, GPU kernels, distributed training; Go, TypeScript, SQL, AWS, Docker, Kubernetes, FastAPI, React; QA & Testing, Spec-driven Development; Statistics, Linear Algebra, HPC

EXPERIENCE

Autonomous Systems Intern – Motional

May 2026 – Aug 2026

- Built the first evaluation metrics and an interactive dashboard to assess multimodal trajectory-prediction models across driving scenarios, surfacing regressions across model versions.
- Built the underlying data pipeline in SQL/AWS Athena and Python (Polars/Pandas) to process large-scale driving logs and surface prediction quality and calibration.

Autonomy & Perception Engineer – CMU Lunabotics

Jan 2026 – Present

- Built the rover's navigation system in ROS2/Nav2, evaluating multiple local planners in simulation before selecting Model Predictive Path Integral (MPPI) control; tuned and validated the full autonomy stack in simulation and on hardware.
- Set up the field test infrastructure (network teleop, on-rover deployment) and trained a segmentation model to identify obstacles from onboard sensor data.

Founding Software Engineer – Tapsta

Jul 2024 – Jun 2025

- Architected end-to-end full-stack social rewards mobile application using React, React Native, FastAPI, and PostgreSQL.
- Designed database architecture with 25+ schemas and optimized client-side operations with caching and debouncing, achieving 30% faster response times. Worked with product team to redesign onboarding flow, driving 40% user retention.
- Orchestrated Plaid API integration alongside REST API creation, establishing secure bank connectivity and enabling automated cashback processing and ACH transfers for over 1,500 application users.
- Made containerized CI/CD pipeline with Docker, AWS, and ArgoCD that reduced deployment time by 70%; led a team of 3 SDE interns, establishing code review processes that contributed to 30% faster feature delivery. Built a semantic people search engine for alumni networks using embedding-based similarity (LangGraph).

Research Assistant - Fan Lab, University of Florida

Feb 2024 – Jun 2024

- Applied genomic models for rare disease prediction using **transformer architectures**, processing **15,000+ DNA sequences**; developed MLP models using protein embeddings (**ESM3**) analyzing RNA-binding proteins.

SDE Intern – Hashira.io

Apr 2023 – Dec 2023

- Taught LLMs to use APIs (incl. function calling), fine-tuning LLaMA/BERT in PyTorch to build a natural-language interface for payroll, achieving 90% natural language to API accuracy and reducing API integration time by 25%.
- Implemented full-stack crypto analytics platform using React, FastAPI, Postgres, Pandas, and Golang with real-time monitoring dashboard and a microservice for a leaderboard system for garden.finance, boosting user engagement by 30%. Improved backend performance using Go and PostgreSQL optimizations, supporting \$150M+ trading volume over 30 days with 20% faster queries.

Research Assistant - Mahindra Ecole Centrale

Dec 2022 – Jan 2025

- Published research on emotion-targeted music generation using FFT, differential evolution, F C-means, self-organizing maps achieving 85% accuracy in classification (IEEE CEC 24). Implemented ALI-GAN model with t-SNE and PCA for clustering analysis on 1000+ samples. Built MIR pipelines extracting 100+ features.

PROJECTS

Blackwell GEMM Kernel & Operator Fusion | TVM/TIRX, C++, CUDA, ML Systems (15-642)

- Built a warp-specialized FP16 GEMM kernel for NVIDIA Blackwell (SM100) with TMA async loads, tensor-core MMA, multi-stage pipelining, and persistent-kernel tile scheduling; added autotuned operator fusion to compile and deploy optimized kernels.
- Reached **1,400 to 1,600 TFLOPs** on saturated matrix dimensions (**75 to 85% of cuBLAS** throughput), validated for correctness against a cuBLAS reference.

SAPS: Step-Aware Pruning Schedule for Diffusion LLMs | PyTorch, LLaDA-8B, ML Systems (15-642)

- Replaced the fixed KV-cache pruning ratio in Sparse-dLLM with an exponentially decaying, step-aware schedule (keep 70% early, 10% late), requiring no retraining and adding one floating-point op per denoising step.
- Cut KV cache from 145 MB to **50 MB (66% less memory)** on LLaDA-8B/GSM8K while matching vanilla accuracy (78.2% vs 78.17%); a Pareto improvement over Sparse-dLLM (31.4% less memory, +1.9pp accuracy). Ran layer-aware and pruning-floor ablations.

Edge-Guided Flow Matching for Controllable Face Synthesis | Diffusion & Flow Matching (10-799) | **Best Poster**

- Built EdgeAdapter, injecting edge features into UNet skip connections and the middle block rather than concatenating at the input, with cycle loss and timestep-weighted conditioning; improved every controllability metric and **halved perceptual distance** (LPIPS 0.34 to 0.17), with the realism tradeoff characterized via ablations.

F1TENTH Autonomous Racing (audited) | ROS2, Simulation, Motion Planning

- Worked through the full F1TENTH lab curriculum in simulation independently; built a local planner in ROS2 for autonomous racing (reactive control + path tracking).

Project RECON | Raspberry Pi, OpenMPI, GlusterFS | \$2000 funding

- Architected 8-node Raspberry Pi 4B **compute cluster** with distributed storage, LDAP authentication, and Slurm job scheduling, serving 400+ students for coursework.

OneAIClick.com | Python, React.js, HuggingFace, PEFT/LoRA (*Built no-code/low code tool*)

- Developed an LLM fine-tuning abstraction tool for HuggingFace models that cut development cycles by 60% and reduced boilerplate code by 75%, enabling rapid idea validation and full data privacy.

GPU-Accelerated Ray Tracer | **CUDA**, Python, NumPy

- Built a real-time GPU ray tracer in CUDA Python, parallelizing rendering across thousands of GPU threads; benchmarked CUDA vs. numba.jit vs. bare Python, achieving order-of-magnitude speedups over CPU-bound implementations.

gitask.org | WebGPU, TypeScript, React | **200+ users**

- Offline, in-browser RAG tool to chat with any GitHub repo via URL swap (github.com → gitask.org). Local LLM inference via WebGPU. Did AST chunking, graph-aware hybrid search, binary quantized embeddings.

LEADERSHIP AND ACHIEVEMENTS

1st Place WaffleHacks 2024 (320+ participants): Built FlashFocus - a Chrome extension to block distractions and turn them into learning with AI flashcard quizzes using RAG system. (React, FastAPI, SQL, Groq)

3rd Place, AWAP 2026 (~80 teams): **Multi-agent** bot (no RL) – Hungarian algorithm for task assignment, Johnson's algorithm + BFS for routing.

1st Place NR Hackathon 2023 (50 teams): Engineered a distributed cluster implementing the Raft consensus protocol from scratch.

Buildspace S5 30+ users - Desktop app, AI sample packs from songs with stem separation, drum extraction & flips for musicians.

President, Enigma, the Computer Science Club (2021-2023): Conducted technical workshops on Gamedev, ML, Linux, heading outreach initiatives increasing the club membership by 40%, reaching over 2000 students. Collabs - Ubisoft, NVIDIA.